Scenario 4C: Strong overdispersion stress test

MSSTNB posterior performance, dispersion recovery, kappa recovery, and oracle diagnostics

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Table of contents

1 Purpose and main conclusion

Scenario 4C is designed as a **strong overdispersion stress test** for the MSSTNB model. It keeps the Scenario 3 mean, exposure, covariate, spatial, and temporal latent-risk structure unchanged, but reduces the negative-binomial dispersion parameter from $r = 15$ to $r = 3$.

The main conclusion is:

> Strong overdispersion substantially increases count variability and upper-tail behavior, but because the overall count level remains high, the model is substantially more stable than in S4A and S4B. Across stable replicates, the model recovers r, latent temporal risk, kappa-level variability, and β_1 reasonably well. However, β_2 remains the more fragile coefficient direction, and one difficult replicate fails through a latent temporal-spatial ridge. Oracle diagnostics confirm that the beta/r block itself is not the source of this failure.

The key evidence is:

1. The data-generation check passes: the average count remains high, the zero proportion remains low, and overdispersion diagnostics increase strongly relative to the reference $r = 15$ setting.
2. The formal fixed-gamma fit gives 9 stable replicates and 1 numerical-instability replicate.
3. Among stable replicates, r is recovered well: the average posterior mean is close to the truth $r = 3$.
4. Kappa-level variability is recovered well in stable replicates, but cell-level kappa recovery remains weak in the difficult oracle replicate.
5. Oracle diagnostics show that fixing λ restores β_2, and fixing both λ and ϕ restores regression and r recovery while eliminating numerical guards.

2 Scenario design

2.1 What is changed in Scenario 4C?

The negative-binomial model is parameterized so that

$$Y_{tj} \sim NB(\mu_{tj}, r),$$

with approximate variance

$$Var(Y_{tj}) = \mu_{tj} + \frac{\mu_{tj}^2}{r}.$$

Thus, smaller r means stronger overdispersion. Scenario 4C changes only the truth value of r:

Table 1: Official S4C strong-overdispersion design.

quantity	value
Scenario ID	S4C_STRONG_OVERDISPERSION_T100
Stress source	Strong NB overdispersion
Reference dispersion	r = 15
Stress dispersion	r = 3
Ratio to reference	0.2
Exposure	Same as S3 reference
Mean structure	Same beta/phi/lambda structure as S3
Gamma	Fixed at 0.8 in fitting
T	100
n1	9
reps	10

The design isolates overdispersion stress. Counts are not globally thinned, and exposure is not reduced or heterogenized beyond the S3 reference structure.

2.2 Difference from S4A and S4B

The distinction among the stress scenarios is important.

- **S4A** lowers the observation-level intensity and creates a global sparse-count regime.
- **S4B** lowers and heterogenizes exposure, creating local information imbalance.
- **S4C** keeps the mean and exposure structure close to S3 but lowers r, creating stronger count variability and heavier upper tails.

Therefore, S4C asks whether the model can separate persistent latent risk from extra-NB observation-level variability when the counts are noisy but still abundant.

3 Data calibration

The S4C data-generation check passed all calibration criteria. The average count remains close to the reference value, while the count coefficient of variation, variance-to-mean ratio, and kappa CV all increase.

Table 2: S4C data-calibration summary across 10 replicates.

metric	value
Mean count	210.595
Zero proportion	0.012
Count CV	1.304
Reference count CV	1.028
CV increase	0.276
Variance/mean	385.153
Reference variance/mean	229.754
Variance/mean increase	155.399
Kappa CV	0.574
Reference kappa CV	0.259
Kappa CV increase	0.314
Maximum count	11,338
Replicates	10

The average count is high and the zero proportion is low. Therefore, S4C is not a sparse-count stress test. Instead, it is a high-variability stress test: count CV increases from about 1.03 to about 1.30, and the variance-to-mean ratio increases from about 230 to about 385.

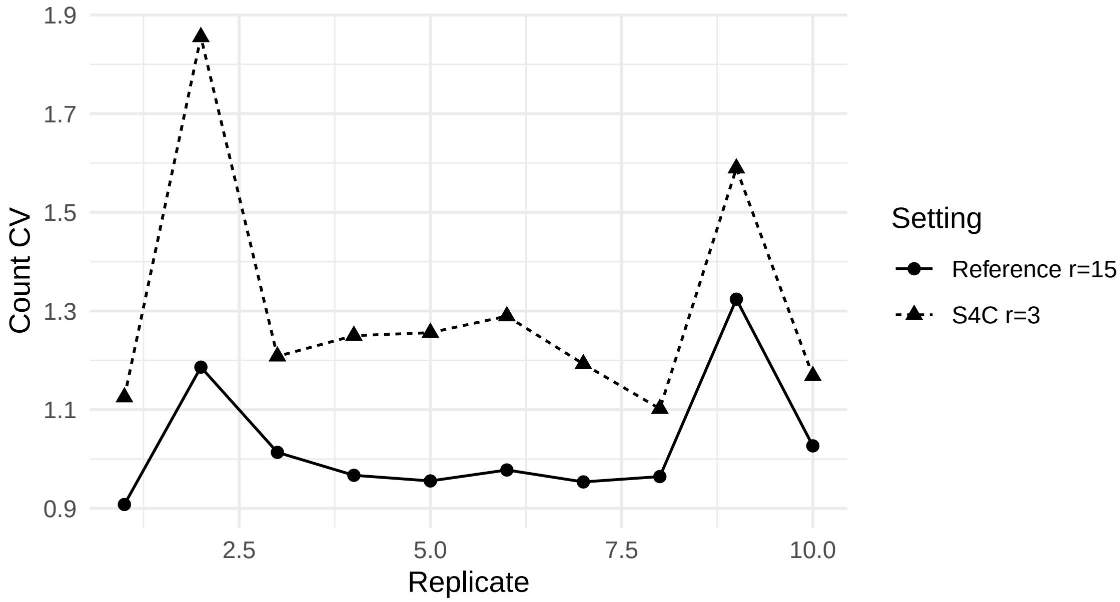

Figure 1: Count CV in S4C and the matched reference setting by replicate.

4 Fitting strategy

S4C is fitted using the same dynamic model structure as S3 and the same fixed-gamma stabilized machinery used in the S4A and S4B diagnostics.

The fitted model estimates

$$\beta_0, \beta_1, \beta_2, \phi, \tau_\phi, r, \kappa, \lambda.$$

The model fixes

$$\gamma = 0.8.$$

The model disables δ and ω updates. This fixed-gamma strategy isolates overdispersion stress rather than re-testing weak identification of γ.

5 Fit-status classification

Each replicate is classified into one of three groups:

- **Stable:** no meaningful numerical guards and no extreme latent-process failure.
- **Soft warning:** small guard activity without severe parameter or latent-process failure.
- **Numerical instability:** large guard activity, extreme β behavior, extreme λ RMSE, or extreme log-λ RMSE.

Table 3: S4C fit-status counts.

status	n_reps	prop
Stable	9	90%
Unstable	1	10%

The result is 9 stable fits and 1 numerical-instability replicate. Thus, S4C is much more stable than S4A and S4B, but it still contains one important failure case.

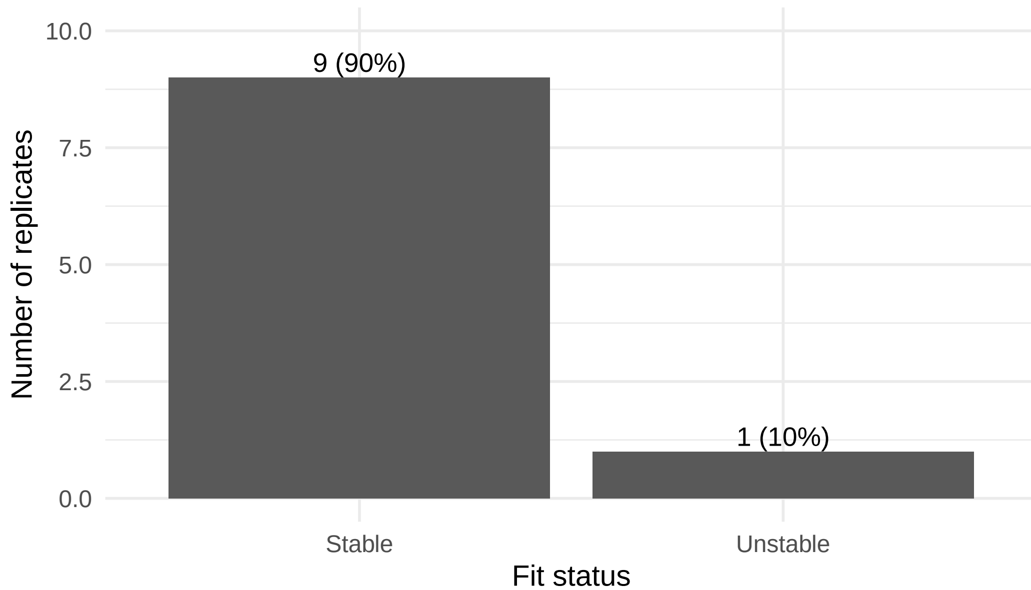

Figure 2: S4C fit-status classification across 10 replicates.

5.1 Replicate-level summaries

The replicate-level results are split into narrow tables to avoid over-wide PDF tables.

Table 4: Replicate-level S4C count and overdispersion summaries.

rep	status	mean	zero	cv	ref_cv	vm	ref_vm
1	Stable	162.5	0.000	1.125	0.908	205.7	134.1
2	Stable	261.8	0.001	1.855	1.186	901.3	362.1
3	Stable	188.0	0.001	1.208	1.013	274.4	187.7
4	Stable	131.2	0.060	1.250	0.967	205.1	124.8
5	Stable	333.7	0.000	1.256	0.956	526.4	303.4
6	Stable	179.6	0.000	1.290	0.978	298.8	164.9
7	Unstable	182.8	0.062	1.193	0.954	260.0	168.2
8	Stable	134.8	0.000	1.102	0.964	163.7	128.0
9	Stable	250.1	0.000	1.589	1.324	631.7	425.7
10	Stable	281.5	0.000	1.169	1.027	384.4	298.7

Table 5: Replicate-level S4C regression, dispersion, and lambda recovery summaries.

rep	status	beta0	beta1	beta2	r	loglam	cor_lam
1	Stable	-1.686	0.512	-0.273	2.895	0.074	0.693
2	Stable	-1.485	0.514	-0.274	3.093	0.057	0.779
3	Stable	-1.752	0.496	-0.590	3.476	0.071	0.591
4	Stable	-2.325	0.498	1.023	3.061	0.139	0.899
5	Stable	-1.387	0.497	-0.580	3.202	0.057	0.704
6	Stable	-1.733	0.504	-0.398	3.133	0.069	0.739
7	Unstable	-2.579	0.532	-4.960	2.793	54.354	0.731
8	Stable	-1.853	0.439	0.102	3.245	0.096	0.809
9	Stable	-1.463	0.505	-0.540	2.885	0.056	0.644
10	Stable	-1.293	0.525	-0.447	3.063	0.056	0.635

Table 6: Replicate-level S4C kappa recovery summaries.

rep	status	truth_cv	post_cv	log_rmse	cor_log
1	Stable	0.593	0.573	0.166	0.968
2	Stable	0.562	0.555	0.141	0.975
3	Stable	0.571	0.550	0.181	0.956
4	Stable	0.577	0.546	0.260	0.909
5	Stable	0.568	0.555	0.126	0.980
6	Stable	0.585	0.568	0.159	0.966
7	Unstable	0.564	1.141	0.462	0.775
8	Stable	0.568	0.546	0.187	0.952
9	Stable	0.595	0.586	0.137	0.977
10	Stable	0.555	0.545	0.122	0.981

Table 7: Replicate-level S4C numerical guard counts.

rep	status	beta_guard	kappa_guard	lambda_guard
1	Stable	0	0	0
2	Stable	0	0	0
3	Stable	0	0	0
4	Stable	0	0	0
5	Stable	0	0	0
6	Stable	0	0	0
7	Unstable	4,373	0	619,350
8	Stable	0	0	0
9	Stable	0	0	0
10	Stable	0	0	0

Replicate 7 is the numerical-instability replicate. It has severe log-λ RMSE, a large lambda-output guard count, and an extreme β_2 posterior mean. It should not be averaged with stable fits as an ordinary recovery result.

6 Posterior performance by subset

Table 8: S4C posterior performance by fit-status subset.

subset	n	mean	zero	cv	beta1	beta2	beta2_sd	r	loglam	cor_lam
All sampled	10	210.59	0.012	1.304	0.502	-0.694	1.575	3.085	5.503	0.722
Stable	9	213.68	0.007	1.316	0.499	-0.220	0.513	3.117	0.075	0.721
Stable+soft	9	213.68	0.007	1.316	0.499	-0.220	0.513	3.117	0.075	0.721
Unstable	1	182.80	0.062	1.193	0.532	-4.960	NA	2.793	54.354	0.731

Among stable fits, β_1 remains accurately recoverable. In contrast, β_2 remains more fragile: its average posterior mean is closer to zero than the truth, and its replicate-to-replicate variability is larger. This continues the pattern seen in S4A and S4B, but with a lower failure rate.

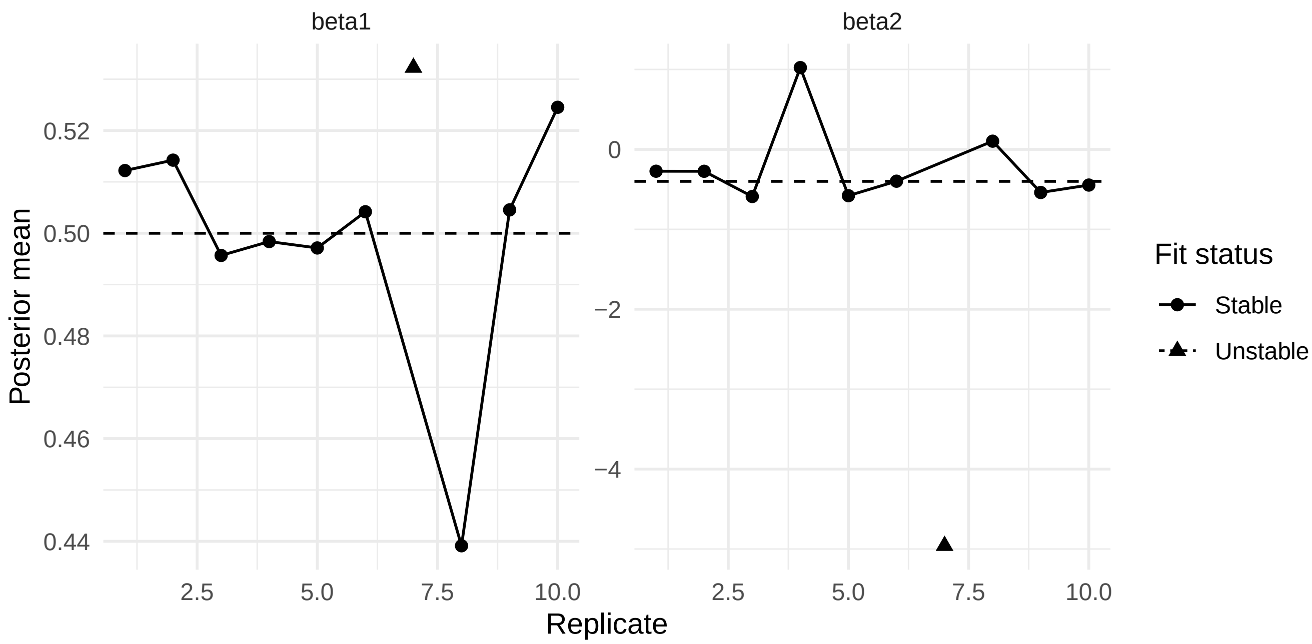

Figure 3: Posterior means for beta1 and beta2 by replicate. Dashed lines show the truth.

7 Dispersion-parameter recovery

Because S4C is a small-r stress test, recovery of r is a central metric.

Table 9: S4C r recovery by fit-status subset.

subset	n	r_truth	r_mean	abs_bias	region_cov
All sampled	10	3.000	3.085	0.170	0.878
Stable	9	3.000	3.117	0.166	0.889
Stable+soft	9	3.000	3.117	0.166	0.889
Unstable	1	3.000	2.793	0.207	0.778

The stable-only average posterior mean of r is close to the truth. This indicates that the model can learn the stronger overdispersion level when the latent temporal and spatial components remain stable.

8 Kappa recovery

S4C also requires checking the gamma-frailty component κ, because small r increases observation-level overdispersion.

Table 10: S4C kappa recovery by fit-status subset.

subset	n	truth_cv	post_cv	cv_bias	kappa_rmse	log_rmse	cor_log
All sampled	10	0.574	0.616	0.043	0.257	0.194	0.944
Stable	9	0.575	0.558	-0.017	0.150	0.165	0.963
Stable+soft	9	0.575	0.558	-0.017	0.150	0.165	0.963
Unstable	1	0.564	1.141	0.577	1.220	0.462	0.775

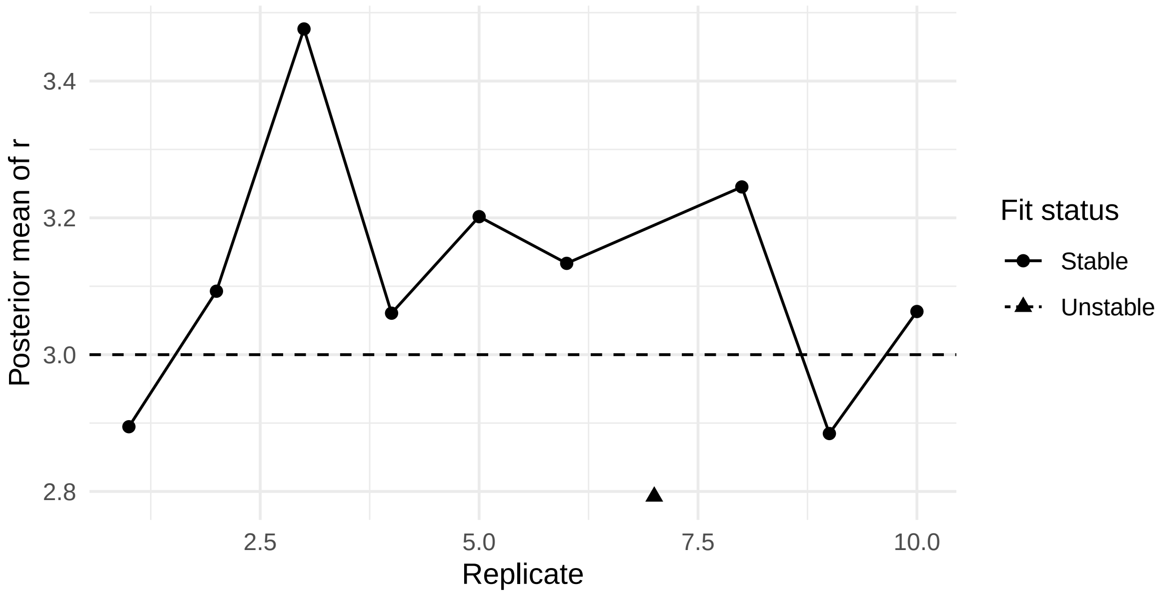

Figure 4: Posterior mean of r by replicate. The dashed line shows the truth r = 3.

Among stable fits, kappa-level variability is recovered well: the posterior mean CV is close to the truth CV, and the average log-kappa correlation is high. The numerical-instability replicate distorts the all-replicate kappa summary and should be interpreted separately.

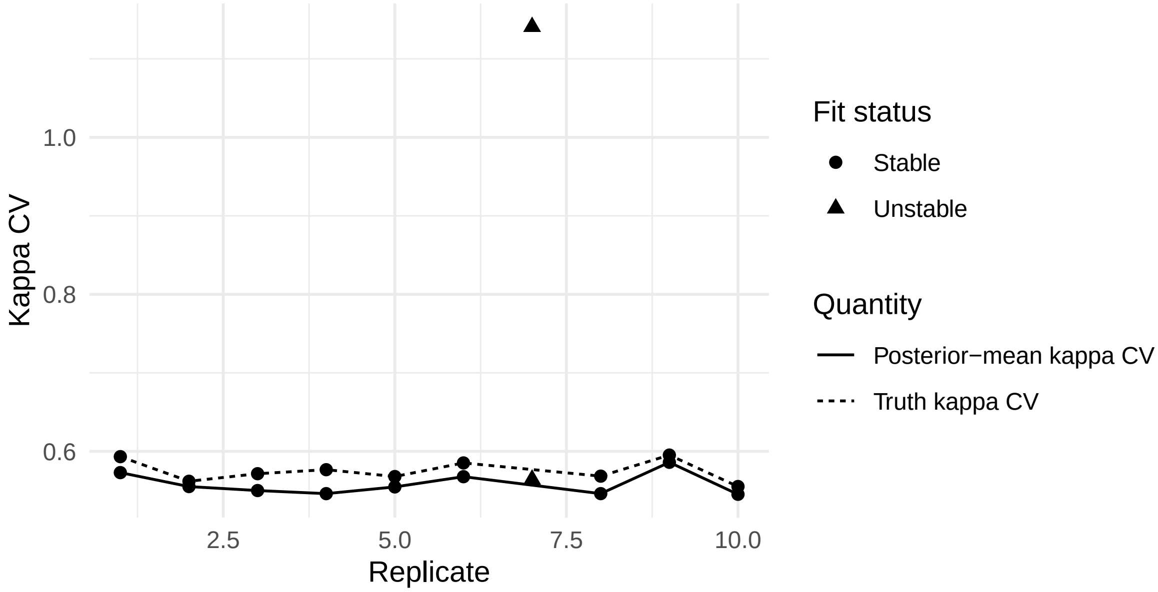

Figure 5: Kappa truth CV and posterior-mean kappa CV by replicate.

9 Comparison with S3, S4A, and S4B

Table 11: Scenario comparison: data scale and instability rate.

scenario	subset	n	mean	zero	instab
S3 control	all control	3	195.05	0.000	0%
S4A sparse	Stable+soft	8	3.17	0.223	20%
S4B exposure	Stable+soft	8	10.32	0.112	20%
S4C overdispersion	Stable	9	213.68	0.007	10%

Table 12: Scenario comparison: beta2 and lambda recovery.

scenario	beta2	beta2_sd	loglam	cor_lam
S3 control	-0.408	0.009	0.047	0.734
S4A sparse	-0.675	2.528	0.206	0.435
S4B exposure	-0.479	1.800	0.185	0.363
S4C overdispersion	-0.220	0.513	0.075	0.721

S4C is very different from S4A and S4B. S4A and S4B are low-information stress regimes; S4C is a high-count but high-variability regime. The S4C stable-only log-λ RMSE is much closer to the S3 control than to the S4A/S4B stress summaries. However, the instability rate is not zero, and β_2 remains the more fragile regression direction.

10 Diagnostic ladder

The diagnostic ladder combines the sampled-lambda S4C fit and the two oracle diagnostics for the failure replicate.

Table 13: S4C diagnostic ladder: step definitions.

step_id	step	n	reps
D1	S4C sampled lambda, stable reps	9	1;2;3;4;5;6;8;9;10
D2	S4C sampled lambda, failure rep	1	7
D3	Oracle lambda, estimate beta/phi/r/kappa	1	7
D4	Oracle lambda + phi, estimate beta/r/kappa	1	7

Table 14: S4C diagnostic ladder: recovery metrics.

step_id	beta2	r	loglam	kappa_cv	log_kappa	cor_kappa
D1	-0.220	3.117	0.075	0.558	0.165	0.963
D2	-4.960	2.793	54.354	1.141	0.462	0.775
D3	-0.487	2.759	0.000	0.668	4.985	0.103
D4	-0.396	3.036	0.000	0.649	4.719	0.120

Table 15: S4C diagnostic ladder: numerical guards.

step_id	beta_guard	kappa_guard	lambda_guard
D1	0	0	0
D2	4,373	0	619,350
D3	0	8,767	0
D4	0	0	0

The ladder shows a clear sequence. In stable sampled-lambda fits, the model recovers r, λ, and kappa-level variability reasonably well. In the failure replicate, sampled λ becomes unstable and β_2 becomes extreme. Fixing λ restores β_2, while fixing both λ and ϕ restores the regression effects and r recovery and removes numerical guards.

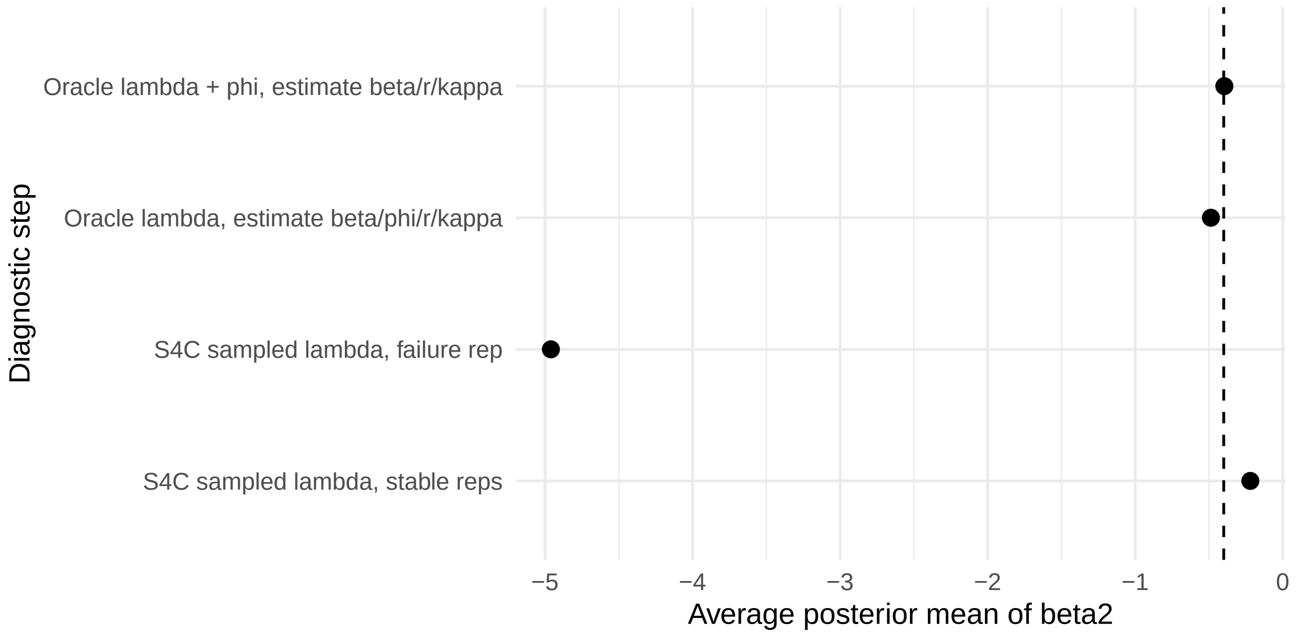

Figure 6: Beta2 recovery across the S4C diagnostic ladder. The dashed line shows the truth.

11 Oracle diagnostics in detail

The oracle diagnostics were run for the numerical-instability replicate, rep 7.

11.1 Step 1: Oracle lambda, estimate beta/phi/r/kappa

In this diagnostic, λ is fixed at the truth, while β, ϕ, r, and κ are estimated.

Table 16: Oracle-lambda diagnostic: regression and dispersion recovery.

rep	beta0	beta1	beta2	r
7	-1.388	0.511	-0.487	2.759

Table 17: Oracle-lambda diagnostic: latent and guard checks.

rep	phi_rmse	loglam	kappa_cv	log_kappa	cor_kappa	beta_guard	kappa_guard
7	19.649	0.000	0.668	4.985	0.103	0	8,767

Fixing λ restores β_2: the posterior mean moves from the sampled-lambda failure value near -4.96 to about -0.49, close to the truth -0.4. However, ϕ and κ remain unstable: ϕ RMSE is large, and kappa guards remain active. This confirms a β_2–λ ridge but leaves a remaining spatial/frailty instability.

11.2 Step 2: Oracle lambda + oracle phi, estimate beta/r/kappa

In this diagnostic, both λ and ϕ are fixed at the truth. Only β, r, and κ are estimated.

Table 18: Oracle-lambda-plus-phi diagnostic: regression and dispersion recovery.

rep	beta0	beta1	beta2	r
7	-8.343	0.508	-0.396	3.036

Table 19: Oracle-lambda-plus-phi diagnostic: latent and guard checks.

rep	phi_rmse	loglam	kappa_cv	log_kappa	cor_kappa	beta_guard	kappa_guard
7	7.61e-13	0.000	0.649	4.719	0.120	0	0

This diagnostic is decisive for the regression and dispersion block. Once λ and ϕ are fixed, β_1, β_2, and r are recovered accurately, and numerical guards disappear. Therefore, the sampled-lambda failure is not caused by a beta/r/kappa implementation problem.

At the same time, cell-level log-κ recovery remains weak in this difficult replicate. Therefore, the correct interpretation is not that every κ_{tj} is perfectly recovered. Rather, fixing λ and ϕ restores the stable regression/dispersion block and removes numerical guards, while cell-level frailty remains hard to recover under strong overdispersion.

12 What strong overdispersion does to the model

12.1 1. It increases variability without creating global sparsity

The average count remains high, and the zero proportion remains low. S4C therefore does not duplicate the sparse-count mechanism in S4A.

12.2 2. It tests whether r and κ can absorb extra-NB noise

The model generally learns the lower r value and recovers kappa-level variability in stable replicates. This is the main positive result of S4C.

12.3 3. It leaves β_2 fragile

Even when the sampler is numerically stable, β_2 remains more variable and more biased than β_1. This matches the pattern seen in S4A and S4B: the β_2 design direction is easier to absorb into latent spatial-temporal effects.

12.4 4. It can still trigger a latent-effect ridge in difficult geometry

Replicate 7 is not simply an extreme-count replicate. It also has difficult identified-scale geometry. Strong overdispersion appears to amplify the instability in that replicate, producing a sampled-lambda failure.

12.5 5. Oracle diagnostics separate implementation error from identifiability limits

The oracle-lambda-plus-phi diagnostic restores β and r recovery and removes numerical guards. This supports the interpretation that the failure is due to joint latent temporal-spatial estimation under difficult geometry, not a generic coding error in the beta/r/kappa block.

13 Practical implications

The S4C results suggest several practical implications for the MSSTNB model.

1. Strong overdispersion is less damaging than sparse counts when information remains abundant. In contrast to S4A and S4B, S4C has a higher stable-fit rate and better latent-risk recovery.
2. Dispersion recovery should be explicitly reported. The posterior mean of r is close to the truth in stable fits, which is an important success for this stress regime.
3. Kappa diagnostics should distinguish variability-level recovery from cell-level recovery. Kappa CV can be recovered well even when cell-level log-κ recovery is weak in a difficult replicate.
4. Fit-status classification remains necessary. Even high-count overdispersed data can produce a failure replicate when the spatial-temporal/covariate geometry is unfavorable.
5. Oracle diagnostics are useful for demonstrating that failures arise from joint latent-effect confounding rather than from the beta/r/kappa implementation itself.

14 Final conclusion

Scenario 4C confirms that reducing the negative-binomial dispersion parameter from $r = 15$ to $r = 3$ creates a strong-overdispersion stress regime without creating global sparsity. The resulting data have high average counts and low zero proportions, but substantially larger count CV, variance-to-mean ratio, and upper-tail behavior than the reference setting.

The fixed-gamma sampled-lambda fit is stable in 9 out of 10 replicates. Among stable replicates, the model recovers the stronger overdispersion level, latent temporal risk, kappa-level variability, and β_1 reasonably well. The main persistent weakness is β_2, which remains more fragile across stress regimes. One difficult replicate exhibits severe sampled-lambda instability. Oracle diagnostics show that fixing λ restores β_2, while fixing both λ and ϕ restores regression and r recovery and removes numerical guards.

Therefore, S4C should be interpreted as a strong-overdispersion stress result with mostly successful recovery under high-count information, plus one informative failure case caused by latent temporal-spatial confounding in a difficult replicate.

15 Recommended reporting language

The following paragraph can be adapted for the manuscript or internal report:

> Scenario 4C evaluated the model under strong negative-binomial overdispersion by reducing the dispersion parameter from $r = 15$ to $r = 3$ while preserving the reference mean, exposure, covariate, spatial, and temporal latent-risk structure. This design increased count variability and upper-tail behavior without creating a global sparse-count regime. Across stable replicates, the model recovered the dispersion parameter, latent temporal risk, kappa-level variability, and β_1 reasonably well. The β_2 direction remained more fragile, and one difficult replicate showed

severe sampled-lambda instability. Oracle diagnostics showed that fixing λ restored β_2, while fixing both λ and ϕ restored all regression effects and r recovery and eliminated numerical guards. These results indicate that strong overdispersion is generally manageable when counts remain high, but can still amplify latent temporal-spatial confounding in unfavorable replicate geometry.